

Sivaganka Sivaneswaran

BSc (Hons) in Information Technology Undergraduate | Full-Stack Developer | MERN Stack | Spring Boot | AI / ML

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Professional Profile

Information Technology undergraduate at SLIIT specializing in full-stack development, backend engineering, and AI-integrated applications. Experienced in building MERN Stack, Spring Boot, React, MySQL, and MongoDB projects with hands-on work in secure authentication, REST APIs, Agile development, and deep learning classifiers. Strong foundation in OOP, data structures, MVC architecture, and scalable web application development.

Core Skills

Core Skills: Full-Stack Development, Backend Development, Frontend Development, REST API Development, MERN Stack, Spring Boot, React.js, Node.js, Express.js, MySQL, MongoDB, JWT Authentication, OOP, MVC Architecture, Agile / Scrum, Git / GitHub, Machine Learning

Education

BSc (Hons) in Information Technology – 2024 – Present

Sri Lanka Institute of Information Technology (SLIIT) – Faculty of Computing

Specialisation: Information Technology

G.C.E Advanced Level (A/L) – MU/Mallavi Central College – Humanities Stream – Completed 2021

G.C.E Ordinary Level (O/L) – MU/Mallavi Central College – Completed 2017

Projects

Online Examination and Result Management System

Tech Stack: Java, Spring Boot, JSP, HTML CSS, Bootstrap, File Handling, GitHub

- Built a full-stack exam system using Spring Boot and Java with modules for user management, exam creation, and automated result processing.
- Applied Linked List and Selection Sort (DSA) for student records and ranking, using a custom file-handling layer instead of a database.
- Designed a responsive UI with JSP, HTML5, CSS3, and Bootstrap for student and admin use.

Blood Donation Management System

Tech Stack: Java, JSP, Servlets, MySQL, Maven, Apache Tomcat

- Built a full-stack web app using Java, JSP, Servlets, and MySQL with role-based access for 6 user roles and secure authentication.
- Implemented modules for appointment scheduling, donor eligibility tracking, and blood inventory management with dashboards and reporting.
- Configured email/SMS notifications for real-time donor alerts and updates.

Multimodal AI System for Visually Impaired (2025)

Tech Stack: Python, OpenCV, NumPy, AI/Deep Learning, Voice Processing, Spring Boot, MySQL

- Built an AI-powered assistive system using OpenCV and Deep Learning for real-time object detection and environmental awareness.
- Developed a data preprocessing pipeline on the COCO dataset covering data cleaning, annotation validation, and class imbalance handling.
- Incorporated a voice interaction module to deliver instant audio feedback, enhancing accessibility for visually impaired users.

AI-Powered Automobile Marketplace (2026)

Tech Stack: MongoDB, Express.js, React.js, Node.js (MERN Stack), TypeScript, HTML, CSS, JavaScript, REST APIs, GitHub

- Built a full-stack MERN marketplace with role-based access (Buyer, Seller, Mechanic,

- Admin) and secure authentication.
- Designed product management, Order System, and Seller Dashboard with dynamic pricing and order tracking.
- Constructed REST APIs and a Rating & Review System using MongoDB, Express.js, React.js, and Node.js.

AI-Powered Automobile Marketplace – Mobile App (2026)

Tech Stack: React Native, MongoDB, Express.js, Node.js (MERN Stack), TypeScript, REST APIs, JWT, GitHub

- Built a full-stack mobile app using React Native, Node.js, and MongoDB with JWT authentication and role-based access (Buyer, Seller, Mechanic, Admin).
- Built product management, Order System, and Seller Dashboard with RESTful API integration and real-time mobile UI.
- Deployed backend on a live platform with a Rating & Review System for full mobile-to-API connectivity.

Certifications

Python for Beginners – University of Moratuwa Verify Certificate	Sep 2024
Web Design for Beginners – University of Moratuwa Verify Certificate	Sep 2024
Microsoft Office – COEL	Sep 2024
Computer Hardware – COEL	Sep 2024
AI Research Exhibition (Think AI Expo 2025), SLIIT	Dec 2025
AI Master Class – LB Camps	Dec 2025

Technical Skills

- Programming Languages:** Java, JavaScript, Python, C, TypeScript
- Core Concepts:** Object-Oriented Programming, Data Structures & Algorithms, MVC Architecture, Agile/Scrum, Unit Testing (JUnit), Database Design, Backend Development
- Frameworks & Technologies:** Spring Boot, React.js, React Native, Node.js, Express.js, RESTful APIs
- Databases:** MySQL (SQL), MongoDB (NoSQL)
- Machine Learning / AI:** Python (Jupyter Notebook), NumPy, Pandas, CNN (EfficientNetB0), Grad-CAM
- Tools:** RESTful APIs, JWT Authentication, Postman, Git, GitHub
- IoT:** ESP32, Sensor Integration (Ultrasonic, GPS, Buzzer), Basic Automation
- Operating Systems:** Ubuntu (Linux Basics)
- Other Skills:** Debugging, Technical Documentation, Time Management, Software Testing, Version Control

Soft Skills

Communication Skills – Team Collaboration – Problem Solving – Technical Communication – Learning Agility – Adaptability – Time Management – Critical Thinking – Leadership

Languages

English
Tamil
Sinhala

Academic References

K.Thiruthanigesan (Senior Lecturer)
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(Pera-Reading) SLIIT,
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The Member of Sri Lanka Young
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